

THE WORLD TECH CHAMPIONSHIP

TECHWIZ 7

Web Application — Software Requirements Specification

CATEGORY: Web & App Development

CookSmart

Theme: Smart Cooking & Recipe Discovery Portal



1.1 Background and Necessity

People often have ingredients at home but struggle to decide what to cook. CookSmart helps users discover recipes according to ingredients, category, cooking time and difficulty, with a simple AI recipe suggestion feature.

1.2 Proposed Solution

The proposed solution is a web application called 'CookSmart'. It will provide the same core concept as the project theme while adapting navigation, layout and interaction to the selected platform.

1.3 Purpose of the Document

The purpose of this document is to present a detailed description of 'CookSmart', including its objectives, scope, website pages, functions and development constraints. It is intended as a reference for stakeholders and developers.

1.3.1 Who Can Use this Document?

Stakeholders and developers of the application.

1.4 Suggested Page / Screen Structure

1. Home
2. About Us
3. Recipes
4. Recipe Details
5. Categories
6. AI Recipe Suggestion
7. Meal Planner
8. Favourite Recipes
9. Cooking Tips / Videos
10. Contact / Feedback

1.5 Scope of Project

The web application will provide the features listed below. The design has already been defined; development should focus on implementing the required functionality and platform-appropriate navigation.

1.6 Constraints

- Develop the portal as a responsive Single-Page Application (SPA).
- Information can be fetched from pre-populated JSON/TXT files; information cannot be written to those files from within the portal.
- Do not use boilerplate or ready-made templates. The student's own design and development skills should be demonstrated.
- AI should be used only for the specified simple AI feature. Students must mention the AI tool/service used and explain its input and output.
- AI-generated visual assets, if used, must be identified in the documentation.

1.7 Functional Requirements

1. Home

Display featured recipes, categories, popular recipes, cooking tips and quick access to AI Recipe Suggestions.

2. User Preference Setup

Accept first name and preferred recipe category.

3. Recipe Library

Display recipe name, image, ingredients, cooking time, difficulty, category and description using sample/JSON data.

4. Recipe Details

Display ingredients, preparation steps, cooking time, servings and cooking tips.

5. Categories

Organize recipes into Breakfast, Lunch, Dinner, Snacks and Desserts.

6. Ingredient Search

Allow users to enter ingredients and find matching recipes.

7. AI Recipe Suggestion

Suggest suitable recipes based on ingredients entered by the user.

8. Search / Filter / Sort

Search and filter by category, cooking time and difficulty.

9. Favourite Recipes

Allow users to save favourite recipes locally.

10. Meal Planner

Provide a simple weekly meal planner.

11. Cooking Tips / Videos

Display cooking tips and videos/images.

12. About Us

Display portal purpose, objectives and team information.

13. Contact / Feedback

Display feedback form and contact information.

1.8 Non-Functional Requirements

- Safe and user-friendly
- Clear and readable interface
- Fast loading and smooth navigation
- Compatible with the target platform/current supported devices
- Consistent visual design
- Responsive/adaptive layout where applicable

1.9 Interface Requirements

Hardware

Intel Core i5 Processor or higher; 8 GB RAM or higher; Color SVGA monitor; 500 GB storage; Mouse; Keyboard.

Software / Technologies

Visual Studio Code or suitable IDE; HTML5, CSS3, JavaScript, Bootstrap (optional), jQuery, Angular/ReactJS, Figma; JSON/TXT data; optional AI API/service.

1.10 Project Deliverables

- Problem Definition
- Design Specifications

- Relevant Flowcharts / Diagrams
- Source Code
- Test Data Used
- Installation / Setup Instructions
- Complete Project Report
- ReadMe file with assumptions
- Mandatory MP4 demonstration video
- Optional hosted/demo build or APK where applicable

AI Usage Note

The AI feature should remain simple and relevant to the selected topic. Students should be able to explain the user input, AI processing/service used and the output displayed in the application.